

HYBRID DEEPFAKE DEFENSE SYSTEM: A DUAL-LAYER APPROACH TO MEDIA AUTHENTICATION AND IDENTITY PROTECTION

Midterm Report
BCS 410-1: Graduation Project I

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Abstract

Generative artificial intelligence has created two separate defensive problems: detecting synthetic media and protecting real individuals from unauthorized facial recognition. Combining both defenses within a single system, however, introduces a conflict that the literature has not previously characterized. Adversarial perturbations applied to real face images for identity protection alter pixel-level statistics in ways that forensic detection models may read as evidence of synthetic origin, causing a legitimately protected photograph to be misclassified as a deepfake.

This report documents the Phase 1 findings of the Hybrid Deepfake Defense System, a dual-layer platform that performs media authentication and identity protection without the two modules working against each other. Three pre-trained forensic detection models were evaluated in inference-only mode on ProGAN data, with cross-domain generalization tested on FaceForensics++. Three adversarial cloaking methods were benchmarked against ArcFace and FaceNet on 1,000 LFW face images. A preliminary conflict analysis measured the Forensic False Positive Rate by passing cloaked images through the top-performing forensic models.

Wang2020 achieves an AUC-ROC of 1.00 on in-distribution ProGAN data; cross-domain performance on FF++ is substantially lower for all three models. PGD-based cloaking methods defeat FaceNet reliably but show limited effectiveness against ArcFace. The conflict analysis confirms that the interaction between the two modules is real and model-dependent: Wang2020 produces an FPR of 11% while UnivFD produces 71% on the same cloaked images. These results directly shape the Phase 2 optimization objectives.

Keywords: deepfake detection, adversarial cloaking, identity protection, forensic image analysis, FaceForensics++, ProGAN, ArcFace, FaceNet, conflict resolution.

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1 Introduction

1.1 Background

Generative Adversarial Networks (GANs), first introduced by Goodfellow et al. in 2014 [1], made photorealistic image synthesis practical by training a generator and a discriminator in adversarial competition. More recently, diffusion-based models, including Stable Diffusion and Latent Diffusion Models [2], have surpassed GANs in perceptual quality and output variety. Synthetic media produced by these systems is increasingly indistinguishable from authentic photographs, even to trained observers.

Alongside generative modeling, adversarial machine learning has produced a separate class of techniques aimed at disrupting AI perception rather than generating content. Adversarial perturbations are structured, often imperceptible modifications to input images that cause deep learning models to misclassify or fail to process them. Applied to face images, tools such as Fawkes [3] use this principle to protect individuals from unauthorized identity extraction by facial recognition systems.

These two research directions, forensic detection of AI-generated media and adversarial protection of real identities, have been developed and evaluated independently. Whether adversarial cloaking tools interfere with forensic detection pipelines has received little attention in the published literature. That gap is the central problem this project addresses.

1.2 Problem Statement

Current defenses against AI-generated media split into two independent tracks. Reactive forensic systems analyze images after the fact to determine whether they were synthetically generated. Proactive cloaking systems modify real images preemptively to stop facial recognition models from extracting biometric identity. No existing platform handles both functions within a single framework.

Each track has its own technical shortcomings. Most forensic detection models were trained on GAN-generated outputs and show measurable accuracy drops on diffusion-based synthetic media, which produces fewer of the frequency-domain artifacts these models were built to detect [4]. On the privacy side, the perturbations that cloaking tools add alter pixel-level and frequency-domain statistics in ways that forensic classifiers may read as evidence of synthetic origin, incorrectly flagging a protected real photograph as a deepfake. This interaction between the two defensive strategies has not been studied systematically.

The Hybrid Deepfake Defense System addresses this gap by building a dual-layer platform that authenticates media and protects identity without the two modules working against each other. Phase 1, documented here, establishes quantified baselines for both modules and takes an initial measurement of the conflict between them.

1.3 Aims and Objectives

The overall aim of this project is to design, implement, and evaluate a dual-layer system that performs forensic media authentication and adversarial identity protection within a unified framework, resolving the operational conflict between the two modules.

Three primary objectives structure the work:

Objective 1 — Forensic Detection: Develop a detection engine capable of classifying images as either authentic or AI-generated. The system will evaluate pre-trained forensic models benchmarked against a standardized dataset. The target performance threshold is a minimum of 90% detection accuracy, measured using Accuracy, AUC-ROC, Precision, and Recall.

Objective 2 — Identity Protection: Develop an adversarial cloaking filter that reduces the success rate of facial recognition to below 20% (Attack Success Rate $\geq 80\%$) while maintaining perceptual image quality (SSIM ≥ 0.90 and PSNR ≥ 30 dB relative to the unmodified source image).

Objective 3 — Conflict Resolution: Verify that adversarial perturbations applied by the privacy module do not cause the forensic detection module to misclassify protected real images as synthetic. This integration test is a formal third objective because its failure renders the combined system unreliable regardless of individual module performance. The passing threshold is a Forensic False Positive Rate (FPR) below 15% across all cloaking method and forensic model combinations.

1.4 Scope

In scope for Phase 1 (this report):

- Inference-only benchmarking of three pre-trained forensic detection models on the ProGAN dataset, with cross-domain evaluation on FaceForensics++ (c23).
- Benchmarking of three adversarial cloaking methods against ArcFace and FaceNet recognition systems on 1,000 LFW face images.
- Preliminary Forensic False Positive Rate measurement using the best-performing cloaking method and the top-two forensic models.

- No model retraining or fine-tuning is performed in Phase 1.

In scope for Phase 2 (Final Report):

- Model selection, threshold tuning, and inference optimization for local GPU deployment.
- Full conflict matrix evaluation and perturbation budget optimization.
- Full-stack web application integrating both modules via FastAPI and React.js.

Out of scope entirely:

- Video deepfake detection; the system operates exclusively on static images.
- Cloud compute, paid APIs, or services with usage limits.
- Audio-based or multimodal deepfake detection.
- Physically-realizable adversarial attacks.

2 Literature Review

2.1 Review of Related Work

2.1.1 Forensic Detection of AI-Generated Images

Wang et al. [5] showed that a ResNet-50 classifier trained exclusively on ProGAN outputs generalizes well across a range of unseen GAN architectures, attributing this cross-generator transferability to low-level frequency artifacts common to most GAN upsampling pipelines. Their model, CNNDetection, remains a widely-used baseline because of this property. Gragnaniello et al. [6] extended this frequency-domain line of work with a patch-based detector trained on full-image GAN outputs, reporting strong performance on older GAN architectures but noting degraded results when the detector is applied to face-centric cropped inputs, where background texture artifacts are absent. Mandelli et al. [7] took a different approach, using noise-residual analysis derived from camera model fingerprinting to identify the characteristic noise patterns introduced during GAN synthesis. This signal is complementary to spatial and frequency detectors, but it similarly depends on GAN-specific artifacts that may not transfer to manipulation-based forgeries.

A limitation shared by all three models is their dependence on GAN-generated training data. Corvi et al. [4] showed that diffusion-based synthetic images, which produce fewer of these artifacts, substantially reduce the accuracy of GAN-trained detectors. Ojha et al. [8] responded to this problem with UnivFD, a universal fake image detector built on a frozen CLIP ViT-L/14 backbone trained with a nearest-neighbor contrastive objective across a broad set of generative models. UnivFD is the only model in this benchmark explicitly evaluated against diffusion-based outputs, making it the most relevant candidate for assessing cross-generator generalization. The FaceForensics++ benchmark suite [9] also contributed an Xception-based binary classifier, fine-tuned on the FF++ dataset across multiple forgery types, which serves as the canonical supervised baseline for face manipulation detection.

2.1.2 Adversarial Identity Protection

Shan et al. [3] introduced Fawkes, which generates perturbations by pushing facial feature representations toward a target identity in the embedding space of a surrogate model. The approach preserves visual quality on higher-resolution inputs but has a documented transfer problem: Radiya-Dixit and Wang [10] showed that Fawkes’ surrogate model does not generalize reliably to independently trained production recognizers such as ArcFace [11] and FaceNet [12], which substantially limits its practical effectiveness.

Cherepanova et al. [13] introduced LowKey, a feature-space attack that explicitly targets robustness to JPEG compression. Komkov and Petiushko [14] proposed AdvHat, a physically-realizable patch attack restricted to the hat region of an image. Shamshad et al. [15] developed CLIP2Protect, which generates perturbations in the semantic embedding space of a CLIP model, attacking recognition at a higher representational level than pixel-space methods. Madry et al. [16] formalized Projected Gradient Descent (PGD) as a principled adversarial attack framework; this provides the theoretical basis for gradient-based cloaking applied directly against a target recognizer in white-box settings.

2.1.3 Facial Recognition Systems

Deng et al. [11] introduced ArcFace, which applies an additive angular margin loss to increase the discriminability of facial embeddings. ArcFace achieves top performance on standard recognition benchmarks and is widely deployed in real-world identity verification systems, making it the primary attack target in this project. Schroff et al. [12] introduced FaceNet, which uses a triplet loss objective to map face images into a compact embedding space suited to verification, recognition, and clustering. FaceNet’s architectural differences from ArcFace make it a useful second target for assessing whether cloaking effects transfer across recognizer designs.

2.1.4 Conflict Between Forensics and Privacy

The interaction between forensic detection and adversarial cloaking has attracted little direct study. Neves et al. [17] noted that adversarial perturbations introduce high-frequency structured noise that resembles the artifacts GAN detectors are trained to flag, suggesting the possibility of systematic false positives. Wang et al. [5] observed that their classifier relies on frequency-domain cues for cross-generator generalization, the same cues that adversarial perturbations also alter. Despite these observations, no published work has directly measured the Forensic False Positive Rate when cloaking methods are applied to real images and the results are passed through deployed forensic detectors. That missing cross-module evaluation is the core gap this project addresses.

2.2 Critical Analysis

The forensic detection literature shares a central limitation: models trained on one generative architecture generalize poorly to others, and this problem is sharpest at the GAN-to-diffusion boundary. Wang2020 and Gragnaniello2021 both perform well within their training distributions but degrade substantially on out-of-distribution forgery types. Mandelli2022’s noise-residual approach is similarly affected by distribution shift, since the noise patterns it targets are specific to GAN upsampling pipelines. UnivFD offers the most credible path to generalization, as its CLIP backbone was trained on a wide set of generative models; however, its performance on face-manipulation forgeries, as distinct from GAN-generated faces, remains

an open question that this benchmark is positioned to answer.

On the privacy side, the tension running through all cloaking methods is between attack success and visual quality. Fawkes and LowKey keep perturbations imperceptible by using surrogate-model-based attacks, but their transferability to target recognizers is limited and worsens as recognition architectures change. PGD-based attacks operate directly against the target model, producing stronger embedding-space disruption at the cost of requiring white-box access. CLIP2Protect attacks at the semantic representation level rather than in pixel space, with transfer properties that are promising but not yet well characterized.

None of the reviewed privacy methods were designed with forensic detection in mind. The perturbations they produce are optimized for recognizer confusion and have unpredictable effects on the statistical properties that forensic classifiers use to distinguish real from synthetic images.

2.3 Gap Identification

Three specific gaps in the existing literature motivate this project.

No benchmark directly compares multiple state-of-the-art forensic detection models under identical evaluation conditions on both in-distribution and out-of-distribution forgery types, using face-centric data with standardized preprocessing.

No published study measures whether adversarial cloaking causes forensic detectors to misclassify protected real images as synthetic. The Forensic False Positive Rate as defined in this project does not appear as a metric in any reviewed paper.

No unified system integrates forensic authentication and adversarial identity protection within a single framework with an explicit conflict resolution mechanism. Existing tools treat each problem in isolation and do not account for their interaction.

3 Solution Determination

3.1 Proposed Solution

The proposed system consists of two cooperating modules: a Forensic Detection Module that classifies uploaded images as authentic or AI-generated, and a Privacy Protection Module that applies adversarial perturbations to real face images to defeat facial recognition. A Conflict Resolution component sits between the two modules and is responsible for ensuring that the privacy module’s output does not trigger false positives in the forensic module.

Phase 1, documented in this report, establishes quantitative baselines for both modules independently and takes an initial measurement of the inter-module conflict. Phase 2 will select the best-performing model from each module, optimize them for local GPU deployment, and integrate both into a FastAPI backend and React.js frontend web application.

All processing runs locally on an NVIDIA RTX 4060 laptop GPU (8 GB VRAM). No cloud compute, paid APIs, or external services are used.

3.2 Evaluation of Alternatives

3.2.1 Alternative 1: Single Forensic Model vs. Multi-Model Benchmarking

Option A – Single pre-selected model: Deploy a single forensic model (e.g., UnivFD) based on published performance claims without independent evaluation.

Option B – Multi-model benchmark (selected): Evaluate multiple models under identical conditions on the same test set before selecting one for deployment.

The benchmarking results reported below directly justify rejecting Option A. Table 3.1 shows results on the primary ProGAN evaluation dataset, and Table 3.2 shows cross-domain results on the FF++ face-crop test set.

Table 3.1: Forensic Model Benchmark Results on ProGAN Dataset (primary evaluation, $n = 1,000$: 500 real, 500 fake). AUC-ROC is the primary ranking metric. *Gagnaniello2021 included for completeness only; excluded from primary benchmark due to FF++ distribution mismatch (see text).

Model	Accuracy	Precision	Recall	AUC-ROC
Wang2020	1.000	1.000	1.000	1.000
Mandelli2022	0.856	0.949	0.752	0.955
UnivFD	0.996	0.992	1.000	0.999
Gagnaniello2021*	1.000	1.000	1.000	1.000

Table 3.2: Forensic Model Cross-Domain Results on FaceForensics++ c23 Dataset ($n = 1,000$: 500 real, 500 fake across four forgery types). All models evaluated in inference-only mode with no fine-tuning. AUC-ROC near 0.5 indicates chance-level performance.

Model	Accuracy	Precision	Recall	AUC-ROC
Wang2020	0.503	1.000	0.006	0.530
Mandelli2022	0.501	0.501	0.980	0.488
UnivFD	0.631	0.892	0.298	0.714

On the ProGAN dataset, Wang2020 achieves a perfect AUC-ROC of 1.00, Mandelli2022 achieves 0.955, and UnivFD achieves 0.999. On the cross-domain FF++ test set, however, results diverge sharply: UnivFD holds at an AUC-ROC of 0.714, while Wang2020 and Mandelli2022 drop to 0.530 and 0.488 respectively, both near chance. A system built on Wang2020 or Mandelli2022 alone, selected without this comparative evidence, would have been incorrectly assumed to generalize to face-manipulation forgeries. Multi-model benchmarking is therefore not a methodological preference but a prerequisite for informed model selection.

Two candidate models were evaluated and excluded before this benchmark was finalized. Gagnaniello2021 [6] was tested using both its ProGAN-trained and StyleGAN2-trained weight variants on the FF++ face-crop test set. Both variants produced logit distributions strongly biased toward the negative class (ProGAN weights: mean logit -7.42 , max post-sigmoid probability 0.56; StyleGAN2 weights: mean logit -5.17 , max post-sigmoid probability 0.47), with AUC-ROC values of 0.572 and 0.578, both near chance. This failure is attributable to a training distribution mismatch: the model was trained on full-image GAN outputs that contain scene-level texture artifacts absent in tightly cropped, RetinaFace-aligned face regions. Gagnaniello2021 was therefore excluded from the primary benchmark, with its ProGAN performance retained in Table 3.1 for completeness. Xception [18] was considered as the canonical FF++-supervised baseline; however, the official model weights distributed through the FaceForensics++ download script were inaccessible during the evaluation window, and weights

from available third-party implementations were found to be unreliable. Xception was excluded from the final benchmark and its absence is documented as a limitation.

3.2.2 Alternative 2: Cloaking Method Selection

Option A – Single established tool (e.g., Fawkes only): Apply a single widely-cited cloaking tool without comparative evaluation.

Option B – Multi-method benchmark (selected): Evaluate multiple cloaking approaches with different attack strategies and compare them on Attack Success Rate, SSIM, and PSNR.

Four cloaking methods were originally considered. LowKey [13] was excluded because its official attack pipeline code is not publicly available; the published repository contains only evaluation scripts, making independent replication infeasible. AdvHat [14] was excluded because it requires hat-region segmentation for physically-realizable patch placement, which is outside the scope of Phase 1’s digital perturbation benchmarking; it was replaced by targeted PGD, which is academically equivalent as a second gradient-based method and gives a direct comparison with untargeted PGD. CLIP2Protect [15] was benchmarked against ArcFace but required approximately 24 hours of GPU runtime for 1,000 images, making full dual-recognizer evaluation impractical within the sprint timeline. CLIP2Protect results are deferred to Phase 2.

The three methods evaluated are:

PGD Untargeted [16]: Maximizes the L2 embedding distance between the clean image and its perturbed counterpart, using an L-infinity norm constraint ($\epsilon = 8/255$, 40 steps). This is a white-box attack implemented directly against ArcFace and FaceNet.

PGD Targeted [16]: Minimizes the embedding distance to a randomly selected decoy identity’s embedding, using the same norm constraint. This is a stronger attack objective than untargeted PGD but requires knowledge of an alternative identity’s embedding.

Fawkes [3]: Generates perturbations using an internal surrogate model to shift feature representations toward a target identity. Included for comparison, with explicit acknowledgment of its documented bypass limitations [10].

Phase 1 results are summarized in Table 3.3. PGD-based methods achieve complete Attack Success Rate against FaceNet ($ASR = 1.00$ for both variants) but fall short of the 80% target against ArcFace ($ASR \approx 0.10$). Both PGD variants narrowly miss the SSIM threshold ($SSIM \approx 0.89$ against the 0.90 target) while meeting the PSNR threshold ($PSNR \geq 33$ dB). Fawkes fails against both recognizers for different reasons. The ArcFace run produces severe image degradation ($SSIM = 0.25$, $PSNR = 14.87$ dB) caused by a resolution mismatch between the 112×112 ArcFace input and Fawkes’ surrogate model design. The FaceNet run produces im-

perceptible perturbations (SSIM = 0.988) that cause no recognizer confusion (ASR = 0.04), consistent with the surrogate transfer failure documented by Radiya-Dixit and Wang [10]. The FaceNet PSNR for Fawkes is undefined because the perturbations were numerically indistinguishable from zero, yielding a mean squared error of zero and an infinite PSNR value; this is a failure mode, not a quality achievement.

Table 3.3: Phase 1 Cloaking Evaluation Results. ASR = Attack Success Rate (proportion of images where recognition failed). SSIM and PSNR measure perceptual quality relative to clean images. Thresholds: ASR ≥ 0.80 , SSIM ≥ 0.90 , PSNR ≥ 30 dB. Values failing their threshold are marked with †.

Method	Recognizer	ASR	SSIM	PSNR (dB)
PGD Untargeted	ArcFace	0.100†	0.891†	33.54
PGD Untargeted	FaceNet	1.000	0.851†	33.04
PGD Targeted	ArcFace	0.098†	0.893†	33.67
PGD Targeted	FaceNet	1.000	0.860†	33.46
Fawkes	ArcFace	0.122†	0.251†	14.87†
Fawkes	FaceNet	0.041†	0.988	—†

† Fails to meet the defined threshold for this metric.

— PSNR undefined: zero MSE indicates no perturbation was applied (surrogate failure).

Baseline recognition: Prior to cloaking, ArcFace achieved a recognition accuracy of 97.50% on the clean LFW corpus ($n = 1,000$) and FaceNet achieved 98.30%, confirming that both recognizers operate at expected performance levels on unmodified images.

3.2.3 Alternative 3: No Conflict Resolution vs. Integrated FPR Gate

Option A – Independent modules, no conflict check: Deploy the forensic and privacy modules as independent tools with no interaction between them.

Option B – Integrated system with explicit conflict measurement (selected): Formally measure the Forensic False Positive Rate and incorporate a conflict resolution mechanism into the deployment architecture.

The preliminary FPR results in Table 3.4 provide direct empirical justification for Option B. PGD targeted was applied to 200 LFW images (selected as the best-performing method by lowest FPR with acceptable quality metrics), and the resulting cloaked images were passed through the two top-performing forensic models. Wang2020 produced an FPR of 0.11 (11%, within the 15% threshold); UnivFD produced an FPR of 0.71 (71%, far exceeding the threshold).

Table 3.4: Preliminary Forensic False Positive Rate (FPR) on 200 PGD-Targeted Cloaked Images. FPR = proportion of cloaked real images classified as synthetic (predicted probability > 0.5). Threshold: FPR < 0.15 .

Forensic Model	Images	Flagged as Synthetic	FPR	Passes (< 0.15)
Wang2020	200	22	0.11	Yes
UnivFD	200	142	0.71	No

A system built under Option A that deployed UnivFD for forensic detection would misclassify 71% of legitimately protected real images as synthetic deepfakes. Conflict resolution is therefore not an optional enhancement but a prerequisite for system reliability.

3.3 Justification

For the forensic module, Wang2020 is selected as the primary deployment model based on its strong in-distribution performance (AUC-ROC = 1.00 on ProGAN) and the lowest FPR in the conflict test (11%). UnivFD is retained as the secondary candidate for scenarios where cross-domain generalization to FF++ manipulation forgeries is prioritized over conflict compatibility. For the privacy module, PGD-based perturbations are selected over Fawkes, whose surrogate transfer failure was both documented in the literature and confirmed empirically here. PGD targeted is the preferred variant, as it achieves equivalent attack success against ArcFace with marginally better SSIM and PSNR than PGD untargeted.

The dual-module architecture is justified by the absence of any published model that simultaneously performs both forensic detection and identity protection. The conflict resolution objective, formalized as Objective 3, is justified by the FPR measurements in Chapter 3 and remains the central technical challenge for Phase 2.

4 Methodology and Requirements

4.1 Methodology

The project follows a phased, research-driven methodology structured around empirical benchmarking in Phase 1 and system integration in Phase 2. Within this structure, an iterative approach consistent with Agile principles is applied: each experimental component (forensic benchmarking, cloaking evaluation, conflict measurement) is treated as an independent sprint with defined deliverables, and findings from one component directly shape the design decisions of subsequent components. This approach is appropriate for a machine learning research project where model behavior on real data regularly deviates from published baselines and where key design decisions (model selection, perturbation budget, conflict threshold) cannot be fixed before empirical evidence is available.

A waterfall-style approach was considered and rejected. A strictly sequential pipeline in which all design decisions are fixed before any code is written would have been inappropriate for Phase 1, where the choice of which forensic model to deploy depends on benchmarking results that do not exist at project inception.

The Phase 1 evaluation design is inference-only. No model retraining or fine-tuning is performed. This constraint is intentional: it isolates the generalization properties of each model as published, produces a reproducible baseline that matches reported results in the original papers, and respects the hardware constraints of local GPU execution.

4.2 Requirements Gathering

Requirements were derived through three complementary approaches. A structured review of the academic literature on forensic detection, adversarial cloaking, and facial recognition provided the technical grounding for both functional and non-functional requirements. Published benchmarks and evaluation protocols from FaceForensics++ [9], the LFW dataset [19], and the original model papers informed the selection of metrics, thresholds, and dataset configurations. Domain analysis of the specific conflict problem, namely the potential for cloaking perturbations to trigger forensic false positives, was carried out through preliminary experimentation that identified the gap documented in Section 2.3 and motivated the formal definition of Objective 3.

4.3 Functional Requirements

- FR-1** The system shall accept a static image as input and classify it as either authentic or AI-generated using a pre-trained forensic detection model.
- FR-2** The system shall output a classification label (authentic or synthetic) and a corresponding confidence score for each analyzed image.
- FR-3** The system shall accept a static face image as input and apply an adversarial perturbation to protect the subject's identity from facial recognition systems.
- FR-4** The system shall return the perturbed image to the user upon completion of the cloaking operation.
- FR-5** The system shall evaluate the perturbed image through the forensic detection module and report whether the cloaked image is classified as authentic or synthetic.
- FR-6** The system shall expose both functionalities (forensic detection and identity protection) through a web-based interface accessible via a standard browser without external service dependencies.
- FR-7** The system shall process all inputs locally on the host machine; no image data shall be transmitted to external servers or cloud services.
- FR-8** The system shall support batch evaluation of multiple images for benchmarking purposes, with results saved to CSV files before any summary computation is performed.

4.4 Non-Functional Requirements

Performance: The forensic detection module shall process a single image in under five seconds on the target hardware (NVIDIA RTX 4060, 8 GB VRAM). The cloaking module shall complete perturbation generation for a single 112×112 face image within 30 seconds using PGD with 40 optimization steps.

Accuracy: The forensic module shall achieve a minimum AUC-ROC of 0.90 on the target evaluation dataset. The cloaking module shall achieve an ASR of at least 80% while maintaining $SSIM \geq 0.90$ and $PSNR \geq 30$ dB relative to the clean input image.

Conflict Constraint: The Forensic False Positive Rate on cloaked real images shall not exceed 15% for the selected forensic model and cloaking method combination deployed in Phase 2.

Reliability: The system shall produce deterministic outputs for a given input image when the random seed is fixed. All random sampling operations shall use seed 42.

Portability: The system shall operate within a single machine with no internet connectivity required after initial model weight downloads. All dependencies shall be managed through isolated Conda environments.

Maintainability: All inference scripts shall follow the project's coding standards: argument parsing via `argparse`, path handling via `pathlib.Path`, GPU memory cleared between model loads via `torch.cuda.empty_cache()`, and results saved to structured CSV files before any in-memory computation.

Privacy: No user-uploaded images shall be stored persistently on the server beyond the duration of the current request. The system shall not log or transmit any biometric data.

5 High-Level System Design

5.1 High-Level Architecture

The system architecture is organized into three tiers: a React.js frontend, a FastAPI backend, and a local model inference layer. Figure 5.1 shows the component structure and data flow.

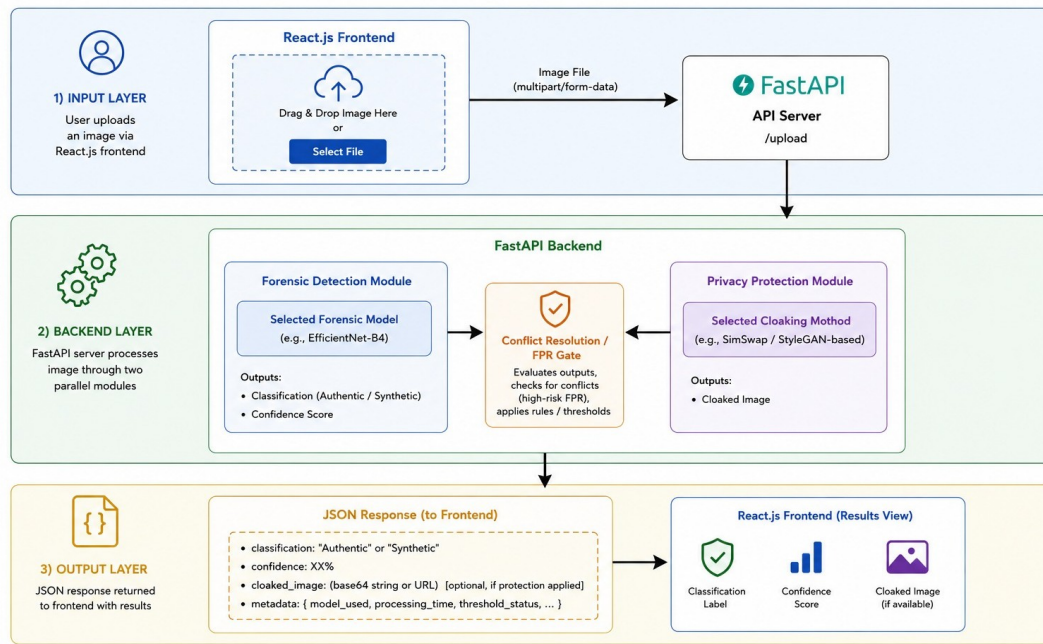


Figure 5.1: High-Level System Architecture of the Hybrid Deepfake Defense System.

The frontend exposes two functions to the user: authenticity verification and identity protection. A user uploads a static image via the browser interface, selects the desired operation, and submits the request. The frontend sends the image to the FastAPI backend as a multipart HTTP POST request.

The backend receives the image, validates the file, and routes the request to the appropriate module. For forensic detection, the image is preprocessed (resized, normalized) and passed to the selected forensic model. For identity protection, the image goes to the cloaking module first, and the resulting perturbed image is then evaluated by the forensic module to check whether cloaking has caused a false positive. The backend assembles a JSON response containing the classification label, confidence score, and, where applicable, the cloaked image encoded in Base64, and returns it to the frontend.

The model inference layer runs entirely on the local GPU. A single model is loaded at any given

time; GPU memory is cleared between model loads to stay within the 8 GB VRAM constraint. This pipeline is illustrated in Figure 5.2 (reproduced from the project synopsis).

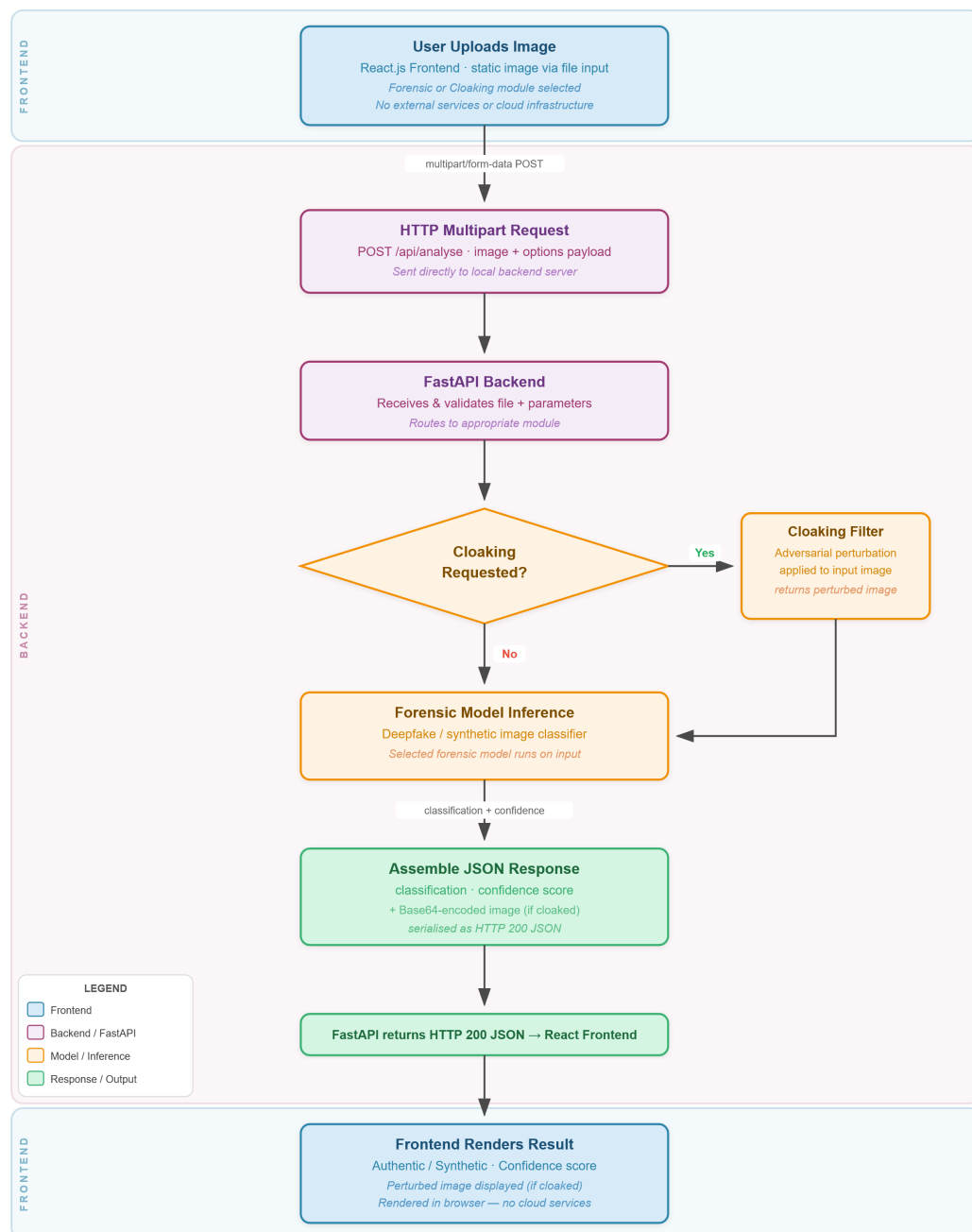


Figure 5.2: Web Application Pipeline Flowchart showing end-to-end request processing.

5.2 Preliminary Prototype

Figure 5.3 shows the preliminary UI wireframes for the web application. Two screens are shown: the upload interface and the results page.

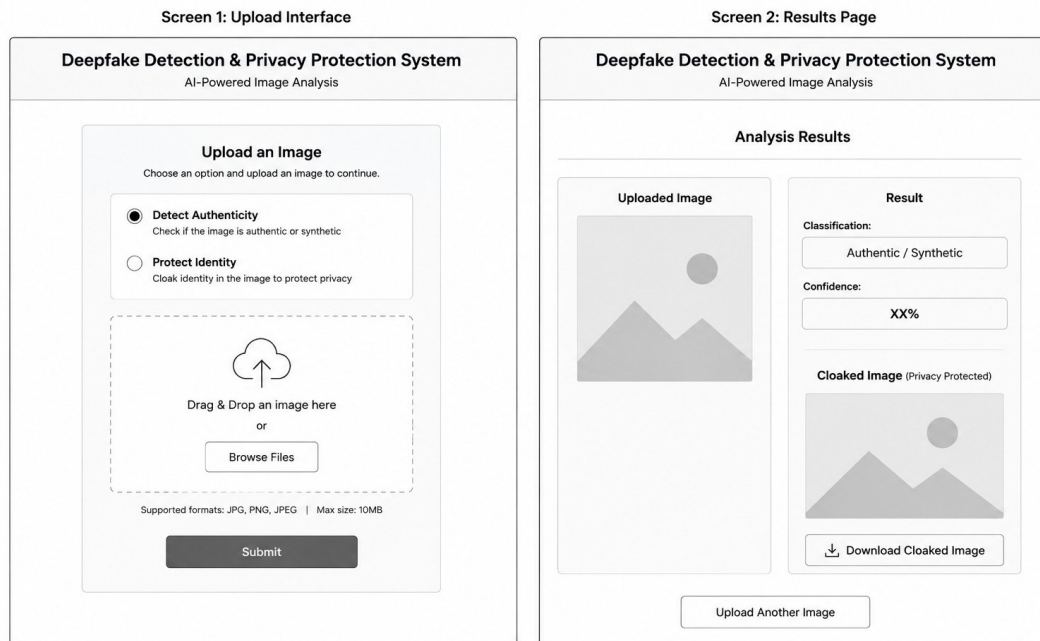


Figure 5.3: Preliminary UI Wireframes: (left) Upload Interface; (right) Results Page.

The upload interface offers two mutually exclusive operation modes. In “Detect Authenticity” mode, the submitted image is analyzed by the forensic detection module and the result is displayed with a confidence score. In “Protect Identity” mode, the submitted image is processed by the cloaking module and the perturbed image is returned for download, alongside a secondary forensic classification of the cloaked image to confirm it is not falsely flagged as synthetic.

The results page is designed to make the conflict metric visible: when a cloaked image is classified as synthetic by the forensic module, the interface displays a warning stating that the selected cloaking method may be incompatible with the deployed forensic model and directing the user to apply a lower perturbation budget.

5.3 Use Case Diagrams

Figure 5.4 shows the use case diagram for the system.

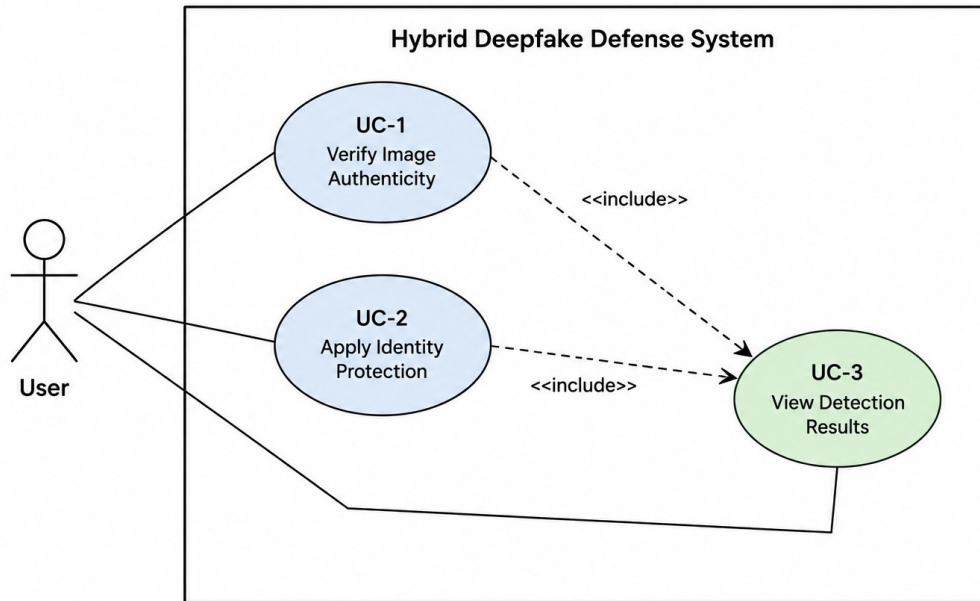


Figure 5.4: Use Case Diagram for the Hybrid Deepfake Defense System.

Three use cases are defined. UC-1 (Verify Image Authenticity) covers the scenario where a user uploads an image and requests a forensic classification. UC-2 (Apply Identity Protection) covers the scenario where a user uploads a face image and requests adversarial cloaking. Both use cases include UC-3 (View Detection Results), which handles the display of the classification output, confidence score, and conflict warning where applicable.

6 Project Management and Conclusion

6.1 Project Timeline

Figure 6.1 shows the Gantt chart for the full project timeline, running from the project proposal submission in April 2026 through the Final Presentation in June 2026.



Figure 6.1: Project Gantt Chart covering Phase 1 (April–May 2026) and Phase 2 (May–June 2026).

6.2 Work Distribution

Table 6.1 lists the primary responsibilities of each team member across both project phases.

Table 6.1: Team Work Distribution by Member and Phase.

Member	Phase 1 Responsibilities	Phase 2 Responsibilities
Mohammad Raafe	Forensics module: all inference code, model benchmarking, GPU execution, results compilation, GitHub maintenance	Phase 2 model optimization, threshold tuning, FastAPI backend integration, system testing
Wagd Haroon	Privacy dataset preprocessing (LFW); report writing and LaTeX compilation	Final report writing; presentation compilation
Saeed Ahmed	Forensic dataset acquisition and preprocessing (FF++); manifest CSV construction	FastAPI backend development; React.js frontend development
Fatima Bey	Privacy module: Fawkes cloaking pipeline, cloaking evaluation metrics, robustness testing	Privacy module optimization; Phase 2 cloaking pipeline

6.3 Conclusion

This report has documented the Phase 1 findings of the Hybrid Deepfake Defense System, a dual-layer platform for forensic media authentication and adversarial identity protection. Chapter 1 established the project’s motivation, objectives, and scope. Chapter 2 reviewed and critically analyzed the relevant literature across four thematic areas – forensic detection, adversarial identity protection, facial recognition systems, and the conflict between the two defense tracks – and identified three concrete gaps that the project addresses. Chapter 3 evaluated alternative approaches across three dimensions: forensic model selection, cloaking method selection, and the decision to include an explicit conflict resolution mechanism, backing each design choice with Phase 1 results. Chapter 4 defined the methodology, requirements gathering process, and a complete set of functional and non-functional requirements. Chapter 5 presented the high-level system architecture, preliminary UI wireframes, and use case model.

The central Phase 1 findings are as follows. Among the three forensic models benchmarked on the ProGAN evaluation dataset, Wang2020 and UnivFD both achieve near-perfect AUC-ROC scores (1.00 and 0.999 respectively), while cross-domain testing on FF++ exposes a sharp generalization gap for all models. Among the three cloaking methods, PGD-based attacks achieve complete Attack Success Rate against FaceNet ($ASR = 1.00$) but fall short of the 80% target against ArcFace ($ASR \approx 0.10$). Fawkes fails against both recognizers due to surrogate transfer limitations, confirming the findings of Radiya-Dixit and Wang [10]. The preliminary conflict analysis shows a model-dependent FPR: Wang2020 produces 11% (within the 15%

threshold), while UnivFD produces 71%, demonstrating that conflict resolution is non-trivial and motivating the Phase 2 optimization work.

Phase 2 will address the remaining open problems: improving ArcFace attack success rate through perturbation budget tuning, resolving the UnivFD conflict through threshold adjustment, integrating the selected modules into a full-stack web application, and running a complete conflict matrix evaluation across all method and model combinations.

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